

MTH204 (Section A)
Graded Worksheet–9 & 10 (4 Marks)

Name:

Roll No.:

Important Notice: The use of electronic gadgets is strictly prohibited during the tutorial. This includes mobile phones, tablets, or any device containing class notes, but anything handwritten is completely allowed.

Q1. Choose the PDE whose solution is $u(x, y)$, where a and b are arbitrary constants:

$$2u = (ax + y)^2 + b$$

Notations: $p = \frac{\partial u}{\partial x}$, $q = \frac{\partial u}{\partial y}$

(a) $qx + py = p^2$

(b) $px + qy = q^2$

(c) $px + qy = p^2$

(d) $qx + py = q^2$

Q2. Choose the PDE whose general solution can be written as below, where f is an arbitrary function:

$$x + y + z = f(x^2 + y^2 + z^2)$$

Notations: $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$

(a) $(y - z)p + (z - x)q = x - y$

(b) $(y - z)q + (z - x)p = x - y$

(c) $(y - x)p + (z - y)q = x - z$

(d) $(y - z)p + (z - x)q = x - z$

Q3. Let $u(x, y)$ be the solution of the given PDE. Then $u(1, 5) = \text{-----}$?

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = 0, \quad \forall (x, y) \in \mathbb{R} \times \mathbb{R}^+$$

$$u(x, 0) = 1, \quad \forall x \in \mathbb{R}$$

Q4. Let $u(x, t) = U(x)T(t)$ satisfy the given 1D wave equation. Then which of the following is true?

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \forall x \in (0, L), t > 0$$

$$\begin{aligned}
u(x, 0) &= f(x), \quad \forall x \in [0, L] \\
u_t(x, 0) &= g(x), \quad \forall x \in [0, L] \\
u(0, t) &= 0, \quad \forall t \geq 0 \\
u_x(L, t) &= 0, \quad \forall t \geq 0
\end{aligned}$$

☐ $U(0) = 0$

☐ $U(L) = 0$

☐ $U'(0) = 0$

☐ $U'(L) = 0$

Q5. If $f(x)$ and $g(x)$ are even functions in the following equation, then which of the following is true about the solution $u(x, t)$ of the equation?

$$\begin{aligned}
\frac{\partial^2 u}{\partial t^2} &= c^2 \frac{\partial^2 u}{\partial x^2}, \quad x \in \mathbb{R}, t > 0 \\
u(x, 0) &= f(x), \quad x \in \mathbb{R} \\
u_t(x, 0) &= g(x), \quad x \in \mathbb{R}
\end{aligned}$$

- It is an even function in x
- It is an odd function in x
- It is not an even function in x
- It is not an odd function in x

Q6. Let $u(x, t) = X(x)T(t)$ be the nontrivial solution of the given equation. Then which of the following is true about k ?

$$\begin{aligned}
\frac{\partial u}{\partial t} &= c^2 \frac{\partial^2 u}{\partial x^2}, \quad x \in (0, L), t > 0 \\
u(x, 0) &= f(x), \quad x \in (0, L) \\
u(0, t) &= 0, \quad u(L, t) = 0, \quad t > 0
\end{aligned}$$

Equation for k :

$$\frac{T'(t)}{c^2 T(t)} = \frac{X''(x)}{X(x)} = k$$

- Nontrivial solution cannot exist when k is non-negative
- k can be a function of x or t
- k can be a complex number

Q7. (Optional)

Derive the D'Alembert solution of the one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

with initial conditions:

$$u(x, 0) = f(x), \quad \frac{\partial u}{\partial t}(x, 0) = g(x)$$